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Research paper

# People do not demand commitment devices because they might not work

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ABSTRACT

Low demand for pure commitment in real markets presents a puzzle. A possible explanation is that individuals are unaware of their present bias and their need for commitment. I test the relationship between perceived intertemporal inconsistency and demand for commitment in a novel context, volunteering. I run an experiment that successfully corrects subjects' beliefs about their present bias and find that this increased awareness does not increase demand for commitment. This low demand for commitment is not driven by a perceived lack of present bias, but rather subjects' accurate belief that they may fail to follow through, even with the offered level of commitment.

## 1. Introduction

Demand for commitment devices is a hallmark prediction of theoretical models of intertemporal inconsistency. However, demand for pure commitment devices in actual markets has fallen short of many behavioral economists' expectations. In the chapter on Intertemporal Choice of the Handbook of Behavioral Economics, Ericson and Laibson (2019) note the following open question: "Why is pure commitment so rare in markets and why is willingness to pay for commitment usually so low (even in lab experiments designed to measure the taste for commitment)?" One suggested explanation for the low take-up of these devices is naivete of present bias. If individuals are unaware that they suffer from intertemporal inconsistency, they will not realize that they need a commitment device to complete their goals. These individuals will, therefore, never demand one. In this paper, I study demand for commitment devices in a novel context, volunteering with a non-profit organization. I analyze an experiment in which I successfully correct subjects' beliefs about their present bias, but find that this increased awareness does not increase demand for commitment. Using the results from this experiment and a theoretical model of commitment demand, I show that the lack of demand does not come from unawareness of present bias. Instead, low commitment take-up is driven by individuals accurately assessing that they may fail to follow through with their volunteering goals, even with the commitment device they have been offered.

I show that willingness to pay for a commitment contract can be broken down into two competing forces. On one hand, individuals who believe that they are present biased realize they need additional incentives to volunteer. This is the classic rationale

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behind commitment devices. However, in settings with uncertainty, commitment devices are also a bet on one's future self to volunteer. I show that, for the studied subjects, the latter force is dominant. Subjects realize they still might fail, causing them to not complete their intended task *and* face the loss associated with the commitment device. In these situations, increasing awareness of present bias only serves to increase this perceived probability of failure, weakening demand for commitment. These results lend further credence to the notion that commitment contracts are not a cure-all for present bias related problems.

Throughout this paper, I define a commitment device as a contract that pays the subject if and only if they follow through and volunteer.[2] To illustrate the promise of these commitment devices, consider the following example. During the week, a present-biased student knows that they would prefer to spend their Sunday afternoon volunteering at a local food bank to watching football. However, when Sunday rolls around, the immediate benefits of watching the game outweigh the heavily discounted future benefit of enjoying the feeling of having helped others that follows driving to the food bank and working hard. This student might be willing to deposit money during the week that they could only get back if they volunteered on Sunday. Then, on Sunday, the increased incentive for volunteering created by this commitment device would tip the scales in favor of volunteering, leading the student to choose the activity they truly preferred.

Decisions about volunteering one's time serve as a novel setting for studying demand for commitment. Volunteering has been shown to be strongly associated with improved happiness and subjective well-being (Aknin et al., 2019); however, the "warm glow" of prosocial behavior often comes after costly effort. This upfront cost and delayed benefit makes the decision to volunteer susceptible to present bias. Pilot surveys showed that there were a large number of individuals who planned to volunteer, yet ultimately failed to do so. This planning-action gap in volunteering was correlated with self-reported procrastination.[3] It is therefore possible that commitment devices would align one's present and future preferences, improving both volunteerism rates and the subjective well-being of individuals.

This paper analyzes a randomized control trial of close to 250 college students who expressed a desire to volunteer. In the experiment, I elicit subjects' beliefs about their probability of following through with their intentions to volunteer during the following month. I also measure their willingness to pay (WTP) for a commitment device that pays them if and only if they volunteer during that month and their actual volunteering activity. One group of subjects receives an information treatment providing them with information on their measured present bias and a model predicted likelihood of volunteering.

The majority of subjects believed they had a 50% or higher likelihood of volunteering in the next month, indicating that many subjects wanted to volunteer and thought they were likely to do so. The information treatment successfully corrects subjects' beliefs about their probability of volunteering. Subjects who are told that they are unlikely to volunteer due to their present bias report that they are less likely to volunteer than similar subjects who were not given any information. Subjects in the information treatment therefore updated their beliefs about their probability of following through towards the experimentally provided information. I find that these subjects who are told that present bias may keep them from volunteering are actually *less* likely to demand commitment. Furthermore, subjects' beliefs about their probability of volunteering are *positively* correlated with willingness to pay for commitment.

I present a theoretical model illustrating how changing a subject's beliefs about their intertemporal (in)consistency and their likelihood of volunteering has opposing effects on willingness to pay for commitment. Decreasing beliefs about the likelihood of volunteering decreases the perceived probability that the individual actually gets paid by the commitment device, decreasing WTP. However, it also increases the perceived need for commitment and decreases the expected cost of volunteering, increasing WTP for commitment. I find that empirically, in this setting, the former effect dominates the latter. As in Carrera et al. (2022), making an individual more aware of their present bias (lowering their perceived probability of following through), lowers their demand for commitment. However, my model also makes predictions about WTP and demand for subsidized commitment. I find that subjects' WTP and demand for subsidized commitment increase with beliefs about probability of following through. The model predicts that if WTP for commitment is increasing in these beliefs, individuals must be uncertain that the commitment device will be successful. This suggests that the subjects recognize the possibility of failing to follow through, even with the offered commitment contract. Results from the experiment and a follow-up survey support the conclusions of the model. 26% of subjects with commitment devices still failed to volunteer. When subjects who did not have a commitment device were asked whether they would have volunteered with the commitment device, fewer than 20% were confident that they would have. Pure commitment contracts are therefore unlikely to be an effective tool for increasing volunteerism or welfare even when individuals are fully aware of their intertemporal inconsistency, as this awareness only suppresses commitment demand. Furthermore, naivete is unlikely to be the explanation for low commitment demand; high potential costs of volunteering are a more likely reason. Even if an individual is fully aware of their present bias, they may not demand pure commitment because there is still a possibility of high costs preventing them from volunteering. This potential for failure, along with the higher expected costs of volunteering, makes commitment an unattractive solution to this problem caused by present bias.

Research on the demand for commitment devices to manage intrapersonal inconsistency of preferences is at a critical juncture. Commitment demand has been studied extensively in experiments. Results surveyed in Schilbach (2019), John (2020), and Carrera

[2] I later distinguish between "pure commitment" and "subsidized commitment". Pure commitment devices exclude contracts that offer the possibility of monetary gain and therefore could only be chosen by intertemporally inconsistent individuals (this definition of pure commitment aligns with the definition of commitment in Bryan et al., 2010). Subsidized commitment devices also help to align incentives for intertemporally inconsistent individuals, but allow for the possibility of monetary gain, and could therefore be chosen by intertemporally consistent individuals as well.

[3] Furthermore, among participants in the main study who stated that they wanted to volunteer in the month prior to the study but failed to do so, a majority indicated that procrastination played a role in their not volunteering.

et al. (2022) show modest demand for commitment across a wide variety of domains impacted by present bias including financial savings (Ashraf et al., 2006), gym attendance (Milkman et al., 2014) and effort at work (Kaur et al., 2010). However, other studies argue that such results are specific to a narrow range of parameters chosen by researchers and are rarely seen in real markets (Laibson, 2015; Ericson and Laibson, 2019). Theoretical models predict that perceived present bias is an important determinant of commitment demand. Given the theoretical prediction that some amount of awareness of one's present bias is necessary for commitment demand, it might be reasonable to suspect that increased awareness of present bias would lead to increased demand for commitment. However, it is possible to come to the opposite conclusion (Carrera et al., 2022). John (2020) shows that measured present bias has no explanatory power for commitment demand, but that beliefs about future temptation are predictive. Zhang and Greiner (2021) on the other hand, find no connection between their proxy for potential sophistication about intertemporal preferences and commitment demand. The direction and magnitude of the effect of informing individuals of their present bias are, therefore, ambiguous and are the empirical question studied in this paper.

This paper studies the effect of learning about one's present bias on commitment demand and thus relates to a literature on awareness of intertemporal consistency. Augenblick and Rabin (2018) use predictions about future choices in a real effort task to estimate substantial naivete over individuals' present bias. Intertemporal preferences are an abstract and complex object making de-biasing experimental subjects challenging. Le Yaouanq and Schwardmann (2022) find that naivete about present bias decreases with experience in a repeated real effort task, suggesting that past behavior may be informative about present bias. Carrera et al. (2022) use subjects' past gym visitation habits and information about other's overestimation of gym attendance to de-bias subjects about their intertemporal inconsistency. This paper implements a similar information intervention and uses subjects' past volunteering behavior and the output of a statistical model to reduce subjects' naivete surrounding their present bias and probability of following through with their goals. This information treatment provides the experimental variation in subjects' beliefs used to estimate the effect of present bias awareness on commitment demand.

This paper is most closely related to Carrera et al. (2022). I find additional evidence, in a novel domain, supporting their conclusion that increased awareness of one's present bias can lead to decreased demand for commitment. Their work and this paper both theoretically predict that demand for commitment is non-monotonic in perceived present bias. This paper is able to use subjects' willingness to pay for a fixed commitment contract to explore where on the demand curve most subjects lie. Using data on subjects' demand for commitment at different prices and additional survey data, this paper is able to further unpack the mechanism behind the effect of information on demand and the low overall commitment demand. Both the willingness to pay data and the follow up surveys suggest that subjects were aware of the possibility of failure even with the offered commitment device. This awareness led to low commitment take-up.

This paper provides evidence that demand for pure commitment to volunteer is quite low. Furthermore, naivete about present bias is not the barrier to demand; thus, improving self awareness would not be helpful. Individuals are aware that their present bias may lead them to fail to volunteer even with commitment.[4] In settings like this, offering pure commitment will be an ineffective and perhaps counterproductive policy tool. Efforts to increase commitment take-up may therefore be misguided.

## 2. Design and implementation

The primary goal of this experiment is to study the effect of beliefs about intertemporal inconsistency on demand for commitment devices. I therefore seek to experimentally affect subjects' beliefs about their present bias and likelihood of following through and volunteering. I then measure subjects' beliefs about their probability of following through, their willingness to pay for commitment, and their actual volunteering behavior. In this section, I describe the design of the experiment, especially the information treatment and the measurement of each of these outcomes.

The main outcome of the RCT is subjects' WTP for a commitment device. In this case the commitment device is a payment of $25 conditional on volunteering in the following month. Willingness to pay is then defined as the size of the unconditional payment that makes the subject indifferent between the unconditional payment and the $25 conditional payment. To measure this WTP for a commitment device, I use the Becker, Degroot, Marschak (BDM) Mechanism (Becker et al., 1964). The BDM mechanism is strategy-proof and thus incentivizes truthful reporting no matter the subjects' risk preferences. The mechanism also allows me to measure the fine-grained WTP for commitment rather than just binary demand at a given price as is measured in many other commitment studies.

In the BDM mechanism, subjects make a series of decisions between two options. They repeatedly choose between an $X dollar unconditional payment or a $25 payment conditional on submitting evidence that they volunteered with a non-profit organization at least once in the next month.[5] Then one of these decisions is chosen at random and implemented. Each of these decisions is analogous to giving the subject $X and asking them whether they would (a) like to deposit that money in exchange for $25 if and only if they volunteer, or (b) deposit the money and receive it unconditionally at the end of the experiment. WTP is then defined as the value $X at which the subject switches from preferring the $25 conditional payment/deposit to preferring the $X unconditional one. Demand for "pure commitment" is a binary measure of whether the individual prefers the $25 conditional payment to a $X payment, when $X \geq 25$. I refer to this as pure commitment because no intertemporally consistent individual would demand such

[4] The possibility of this commitment device failure is presented in Bai et al. (2021) and John (2020). This study suggests that individuals are aware of this possibility and make decisions regarding commitment accordingly.

[5] Subjects are actually told that the $X unconditional payment is conditional on uploading any photo at any time during the same month. This was to alleviate concerns that low commitment demand might be due to the effort costs or timing of the documentation process of the commitment device.

a contract. Demand for subsidized commitment refers to the binary measure of whether the individual prefers the $25 conditional payment to a $X payment, when $X < 25$. I denote such demand as "subsidized" to distinguish it from the pure commitment devices often studied by the literature. It is possible that an intertemporally consistent individual would choose such a commitment device with no commitment related motives. In the results, I will specifically focus on demand for pure commitment where $X = 25$ and demand for subsidized commitment where $X = 24$.

To measure the effect of information on this WTP, an information treatment group is given information about their present bias as measured by the first stage of the study.[6] While subjects' intertemporal preferences were measured over monetary payments, our calibration study found a significant relationship between this measure and volunteering behavior. Subjects are told whether our model predicts that they are present biased, and if so, whether our model predicts that they will struggle with procrastination in volunteering.[7] I then measure differences in demand for commitment between the control and information treatment groups.[8]

Given this information treatment, it is important to measure the change in subjects' beliefs about their present bias. The goal of the information treatment is to correct subjects' beliefs about their present bias and probability of following through. To proxy for subjects' beliefs about their own intertemporal consistency, subjects are asked to state the probability that they will volunteer in the next month. While this probability is, of course, the function of many factors, all subjects have indicated that they wish to volunteer next month, making this a belief about how their future self will follow through on current desires. These beliefs are elicited after the information treatment, but before any mention of commitment devices. I report the effects of the information treatment on these beliefs as well as the relationship between these beliefs and commitment demand.

To measure the effect of commitment devices on volunteering decisions, a treatment group receives their choice of an unconditional payment and the commitment device from the BDM mechanism. The control group receives an unconditional payment as long as they upload a photo of themselves. All subjects who received the unconditional payment were asked to self report whether or not they volunteered (and were paid $5 for completing the follow up survey). Subjects who were selected for the commitment device are asked for any photo evidence that they volunteered. To reduce the impact of forgetfulness on the volunteering rates of each group, subjects in both groups receive multiple emails reminding them that they are taking part in the study and that they had expressed interest in volunteering. All subjects are also asked additional questions about their decision making process.

Fig. 1 outlines the overall experimental design and the timing of the components described above. The design of the study was pre-registered in the AEA RCT Registry as study AEARCTR-0007315 and was approved by the IRB at Boston College. Subjects were undergraduate students recruited via email. All study materials were administered via Qualtrics.[9]

## 3. Experimental results

Data collection began in March 2021 when 346 undergraduate students completed the preliminary survey. Of the initial subjects, 249 indicated that they were interested in volunteering with a non-profit organization during the next month (making them eligible for the remainder of the study) and completed the commitment demand elicitation portion of the study. 228 subjects completed the follow up survey one month later in which volunteering activity was measured.[10] Information treatment groups were balanced across observable characteristics.[11]

### *3.1. Beliefs about probability of volunteering*

Subjects' beliefs about their probability of volunteering in the next month were generally overconfident but predictive of actual volunteering. On average, subjects believed that they had a 48% probability of volunteering in the next month, which is greater than both the 32% average model predicted probability and the 20% actual follow through rate. Both subject beliefs and model predicted probability are predictive of actual volunteering, but the model predicted probability is more incrementally predictive.[12]

[6] Subjects were paid $5-8 to complete an initial intake survey. The amount and timing of their payment was based on incentivized questions about allocation of payment over time which were used to estimate present bias. Monetary payments are an imperfect proxy for intertemporal preferences over effort. However, importantly, this measure is incrementally predictive of future volunteering behavior. See Appendix Table 2.

[7] A probit model of volunteering behavior was fit prior to the beginning of the study based on a preliminary survey of 87 students at the same institution. Importantly, in this preliminary survey, both past volunteering behavior and measured present bias over monetary payments were significantly predictive of volunteering activity. This model is then used to predict subjects' probability of volunteering which is then converted to a qualitative statement about their likelihood ("very likely", "likely", "somewhat likely", "somewhat unlikely", "unlikely", "very unlikely"). See Appendix Table 2 for details about the predictive model. Appendix Figure 5 shows the information treatment as it is displayed to subjects.

[8] Many efforts were made to reduce concerns about experimenter demand effects or image signaling (see Andreoni and Serra-Garcia, 2021 for analysis of image signaling and intertemporal inconsistency and Ariely et al. (2009) for image concerns and monetary incentives) affecting commitment demand. The study did not involve any direct interaction with the experimenter and subjects were assured that all of their responses would be anonymized before analysis. However, if social image concerns remained, they should affect both the treatment and control groups. Therefore, this analysis of the difference in demand between the control and treatment groups should not be affected by demand or social image effects.

[9] Relevant study materials can be found in the Appendix. All are available upon request.

[10] Results are reported on the 249 subjects who completed the belief elicitation and commitment demand portions of the study. The 21 subjects who did not complete the follow up are assumed to have not volunteered during the month. Qualitatively similar results are obtained using the subset of 228 subjects that completed the follow up.

[11] Demographics and summary statistics for each variable collected by information treatment group are presented in the appendix in Table 4

[12] See Table 3 in Appendix for regressions of actual volunteering on model predictions and subject beliefs.

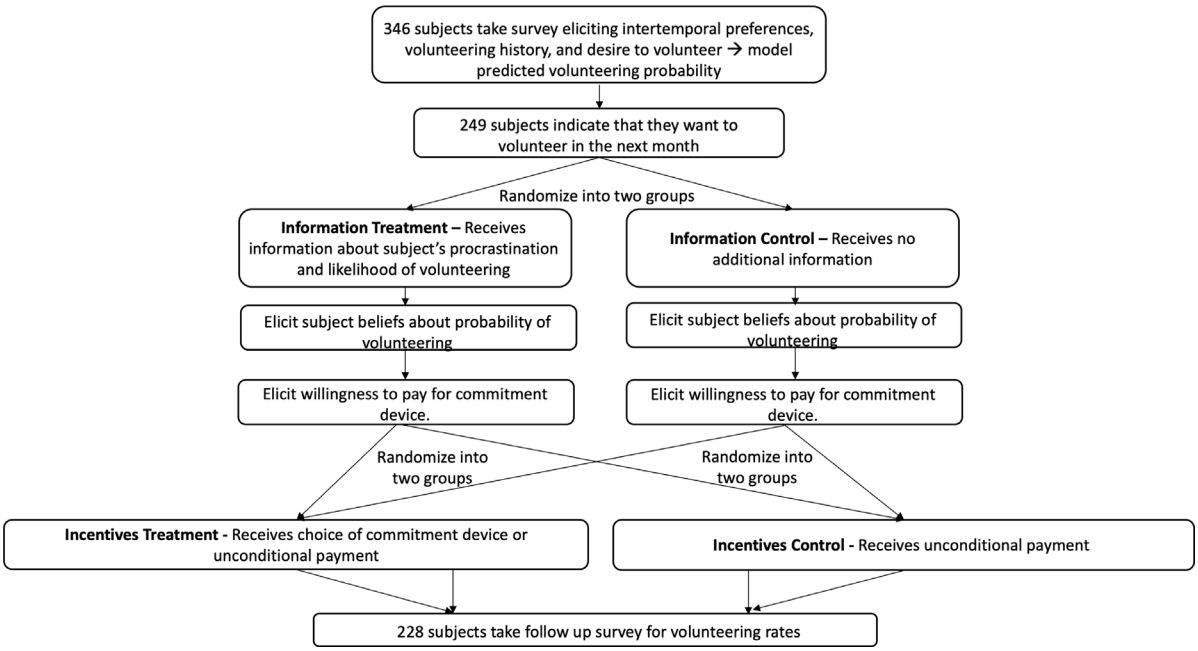



**Fig. 1.** Experimental design and timeline.

As intended, the information treatment moved subjects' beliefs towards the experimentally provided information. In the information treatment, subjects are shown a qualitative statement about their probability of volunteering as predicted by the probit model fit with pilot study data. Subjects' beliefs are, of course, correlated with this model predicted probability, whether the subject was shown the information treatment or not. However, subjects' beliefs are closer to the model predictions in the information treatment than in the control group. For example, subjects who the model predicted were very likely to volunteer that were then told that they were very likely to volunteer had higher beliefs about their likelihood of volunteering than subjects who the model predicts were very likely to volunteer but were told nothing (and vice versa for subjects who were very unlikely to volunteer). Subjects who were in the information treatment had more accurate beliefs than those in the control group. To see this, we compare beliefs of subjects with the same model predicted probability of volunteering (Predicted $\text{Prob}_i$ in the equation below) who have been told their model predicted probability (received the information treatment; $\mathbb{1}(\text{Info Treatment}_i) = 1$ in the equation below) and those who have not. A regression estimating the equation,

$$\text{Beliefs}_i = \beta_1 \text{Predicted Prob}_i + \beta_2 \mathbb{1}(\text{Info Treatment}_i) * \text{Predicted Prob}_i + \beta_3 \mathbb{1}(\text{Info Treatment}_i) + \epsilon_i, \tag{1}$$

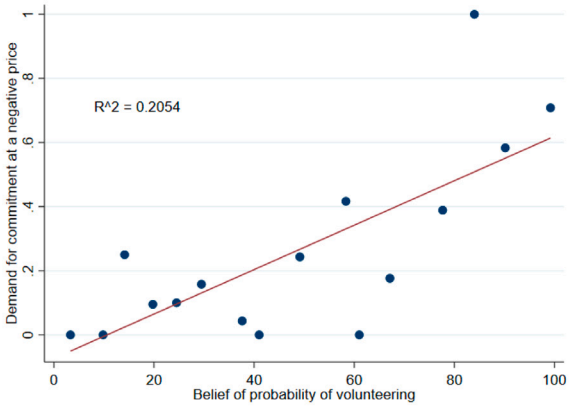
finds a significant positive coefficient on both the model predicted probability of volunteering ($\beta_1 = 0.44$, $p = 0.000$) and the model predicted probability if they were shown it ($\beta_2 = 0.19$, $p = 0.046$). See Table 1 Column 1 for regression results. The positive coefficient on the interaction term ($\beta_2$) suggests that the information treatment had the intended effect on subjects' beliefs.

These results demonstrate that the information treatment was effective at changing subjects' beliefs and reducing their naivete. Subjects clearly attended to the information treatment and updated their beliefs accordingly. This experimental variation in beliefs can then be used to study the effect of beliefs on WTP for commitment.

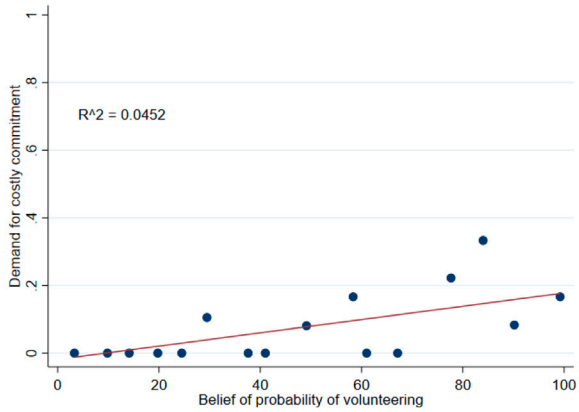
### 3.2. Demand for commitment

Given the reduced naivete among subjects, one might expect increased commitment demand. Instead, the results in this section indicate that overall commitment demand was low and that demand was *increasing* in subjects' beliefs about their likelihood of following through.

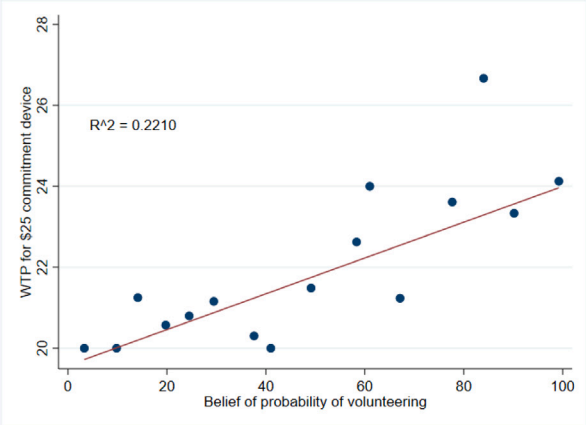
Of the 249 subjects who indicated that they would like to volunteer, 65 (26.1%) chose a $25 payment contingent on volunteering and uploading evidence of doing so over a $24 unconditional payment (subsidized commitment demand). However, when asked to choose between a $25 conditional payment and a $25 unconditional payment, only 19 (7.6%) chose the commitment device (pure commitment demand). Demand for commitment at both of these prices is positively correlated with individuals' beliefs about their ex-ante probability of volunteering (see Fig. 2). Estimates from a simple linear regression predict that a 10% increase in a subject's belief that they will volunteer next month corresponds to a 7% ($p = 0.000$) increase in the probability that they chose subsidized commitment and a 2% ($p = 0.001$) increase in the probability that they chose pure commitment, although beliefs are far more predictive of subsidized commitment than pure commitment ($R^2$ of 0.205 vs 0.045). WTP for commitment is also increasing in the




(a) Bin-scatter of subsidize commitment vs beliefs about likelihood of volunteering. The y-axis represents the fraction of subjects demanding subsidized commitment (choosing a $25 dollar commitment device over a $24 certain payment). The x-axis represents subjects' stated probability of volunteering in the absence of commitment.




(b) Bin-scatter of subsidize commitment vs beliefs about likelihood of volunteering. The y-axis represents the fraction of subjects demanding pure commitment (choosing a $25 dollar commitment device over a $25 certain payment). The x-axis represents subjects' stated probability of volunteering in the absence of commitment.




(c) Bin-scatter of WTP vs beliefs about likelihood of volunteering. The y-axis represents willingness to pay for a $25 commitment device in dollars (the subject's switching point at which they prefer a $25 commitment device vs. a certain payment). The x-axis represents subjects' stated probability of volunteering in the absence of commitment.

**Fig. 2.** Subject beliefs and commitment demand.

subject's perceived probability of volunteering. A simple linear regression predicts that a 10% increase in a subject's belief that they will volunteer next month corresponds to a $0.44 ($p = 0.000$) increase in WTP for the $25 commitment contract ($R^2 = 0.2210$).

There are possibly unobservable subject characteristics that are correlated with both subjects' reported beliefs and their demand for commitment that could bias these estimates. Our information treatment allowed us to experimentally manipulate subjects' beliefs and separate these effects. I run a linear regression of demand for subsidized commitment on the signal that I show subjects, the

**Table 1**
Effects of information treatment.

| | (1) Belief of Likelihood of Volunteering | (2) Demand for Subsidized Commitment | (3) Demand for Pure Commitment | (4) WTP |
|---|---|---|---|---|
| Predicted Prob_*i* | 0.439*** | 0.003*** | 0.000 | 0.021*** |
| | (0.068) | (0.001) | (0.001) | (0.008) |
| $\mathbb{1}$(Info Treatment_*i*) | 0.190** | 0.003* | −0.000 | 0.008 |
| ∗ Predicted Prob_*i* | (0.095) | (0.002) | (0.001) | (0.011) |
| $\mathbb{1}$(Info Treatment_*i*) | −9.078** | −0.027 | 0.013 | −0.091 |
| | (4.023) | (0.071) | (0.046) | (0.449) |
| Constant | 37.319*** | 0.132*** | 0.064** | 21.024*** |
| | (2.823) | (0.050) | (0.032) | (0.315) |
| Observations | 249.000 | 249.000 | 249.000 | 249.000 |
| $R^2$ | 0.353 | 0.143 | 0.001 | 0.089 |

Standard errors in parentheses.
Table notes: Predicted $\text{Prob}_i$ refers to the model predicted probability that the subject $i$ will volunteer. $\mathbb{1}(\text{Info Treatment}_i)$ is a binary variable that is 1 if the subject was in the information treatment and was shown the model predicted probability. The interaction term captures the model predicted probability *if* it was shown to the subject. Column 1 reports the estimated coefficients when the outcome variable is the subjects reported likelihood of volunteering. The outcome variable in Column 2 is whether the subject chose a $25 commitment device over a $24 unconditional payment. In column 3, it is whether the subject chose a $25 commitment device over a $25 unconditional payment. In column 4, it is the subjects' WTP (in dollars) for a $25 commitment device.
* $p < 0.10$.
** $p < 0.05$.
*** $p < 0.01$.

model predicted likelihood (the signal that I *would show* subjects if they were in the treatment group), and a dummy variable for the information treatment (See Table 1, Column 2). This regression estimates the same model as Eq. (1) but with demand for a $25 commitment device at a $24 price as the dependent variable. Specifically, I estimate

$$\text{Demand}_i = \beta_1 \text{Predicted Prob}_i + \beta_2 \mathbb{1}(\text{Info Treatment}_i) * \text{Predicted Prob}_i + \beta_3 \mathbb{1}(\text{Info Treatment}_i) + \epsilon_i, \tag{2}$$

This gives me a point estimate that for every 10% increase in the probability signal that subjects in the information treatment *actually received*, they were 3.3% more likely to choose subsidized commitment. Running the same regression on demand for pure commitment (See Table 1, Column 3), all coefficients are indistinguishable from zero, (in fact, the point estimate of the coefficient on the signal shown to subjects is negative). Finally, running the same regression but with WTP as the dependent variable, produces a positive but statistically insignificant coefficient on the signal that subjects receive (Table 1, Column 4). Measurement error due to left censoring of this variable is likely to account for the lack of an effect. Subjects were not asked about any price below $20, and 168 subjects (67%) had a WTP less than or equal to this price. For this reason, demand for commitment at different prices is the preferred measurement.

Overall, it is clear that the information treatment had an effect on both beliefs and demand for commitment. Individuals who believed they were less likely to follow through were less likely to demand commitment.

### *3.3. Effect of commitment on volunteering*

Given the low take-up rates for the commitment device, the aggregate effects of offering commitment to subjects on volunteerism were minimal. Of the 139 subjects randomly selected into the incentives treatment, 23 had a WTP above their randomly selected price and were thus paid according to the commitment device. The treatment effect of offering commitment was indistinguishable from zero. Average volunteering rates across treatment groups were 20%.[13]

Measuring the effect of the commitment device on the volunteering rates of those who had it is more difficult because selection into commitment was not random. Unconditional average volunteering rates for those with the commitment device are 59 percentage points higher than those without ($p = 0.000$). We can partially control for these selection issues by controlling for the subjects WTP by adding it to the regression. Conditional on the subject's WTP, selection into the commitment device is random. Accounting for this selection, the treatment effect of a commitment device on those who had one is a 43 percentage point ($p = 0.000$) increase in volunteerism. This is a sizable increase in probability, indicating that commitment devices were fairly effective for those who had them; but notably, volunteering rates were not perfect among those who had commitment (74%). Alternatively we could control for selection by adding the model predicted volunteering rate to the regression. Doing so gives us a treatment effect on the treated of 40 percentage points ($p = 0.000$). This regression is visualized in the bin-scatter plot of Fig. 3, confirming that the commitment contract greatly improved follow-through, but considerable uncertainty remained.

[13] The information treatment also had no effect on volunteering rates.

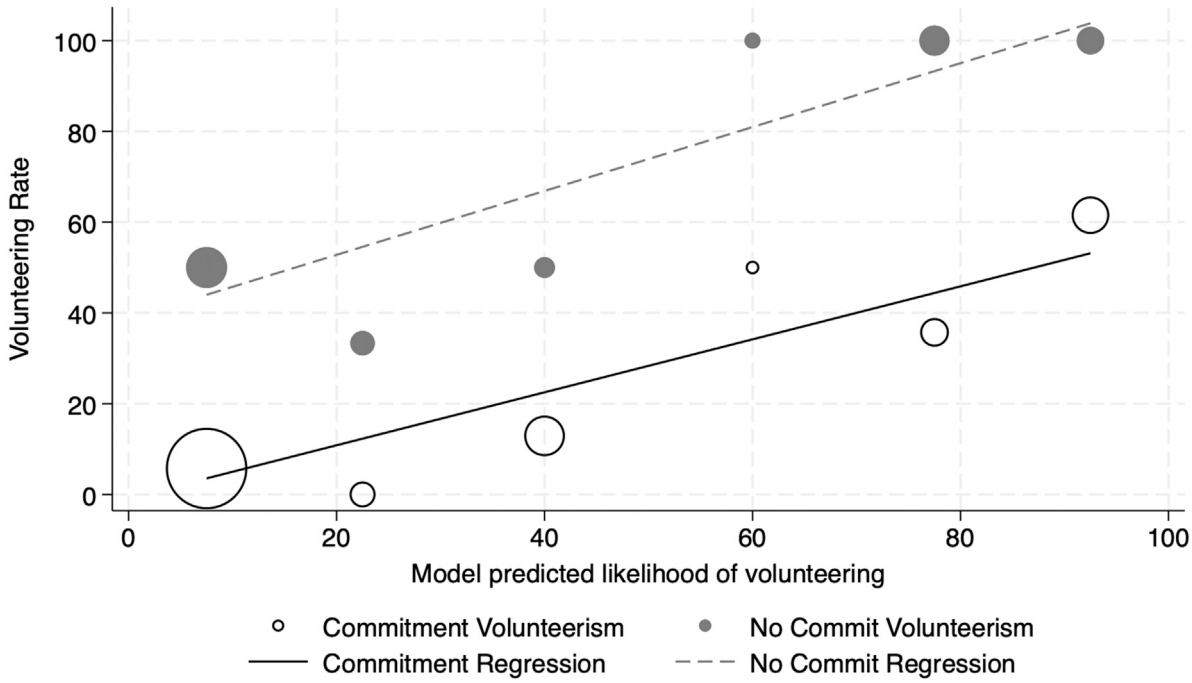



**Fig. 3.** Bin-scatter of volunteering rates for subjects with and without commitment devices. The y-axis represents the actual rate of volunteering in the month of the study. The x-axis represents the model predicted volunteering rate. Data plotted in gray refers to those who had commitment devices, while data in white refers to those who did not. The size of the points represents the number of subjects represented by each point. Lines of best fit are also plotted.

## 4. Theoretical model and discussion

A commitment device allows an individual's present self to increase the incentive for their future self to take an action. This alters the present self's expected future utility through three channels. First, it increases the probability that the future self takes the action (volunteering in this study), increasing their ex-ante expected utility. However, it also introduces the possibility that the individual will still fail to volunteer, additionally causing them to lose the money they put at stake. It also increases the individual's expected costs of volunteering by causing them to volunteer in higher cost situations. If an individual chooses to implement a commitment device, it must be that they believe the first effect is greater in magnitude than the latter two effects.

The model below outlines conditions on the individual's beliefs under which demand for commitment is increasing, decreasing, or unchanging in the individuals perceived probability of following through. In Proposition 1, I find that WTP for commitment is increasing in this belief if and only if they are not 100% certain that they will volunteer, even with the help of a commitment device. If individuals think that commitment might fail, then the effect of their beliefs on the probability of losing the money dominates. Proposition 2 states that WTP for commitment is decreasing in this belief if and only if they are 100% certain to volunteer with a commitment device, but uncertain without. If individuals think that commitment is sure to work, then the effect of their beliefs on their need for commitment dominates. Proposition 3 simply states that an individual who is certain that they will volunteer will always have a WTP of \$25 for a \$25 commitment device. Finally, Proposition 4 demonstrates that there exist scenarios in which individuals have a positive expected value from taking an action, yet they would never demand commitment, no matter their beliefs.

### 4.1. Model

Consider an individual who makes a series of decisions over three periods $t = 0, 1, 2$.

1. $t = 0$: The individual evaluates their probability of volunteering, $\pi_{i,no}$, and chooses between a commitment contract (described below) and an unconditional payment.
2. $t = 1$: The individual realizes a cost shock $c_i$ from the known distribution $F_i(c_i)$ and chooses whether or not to volunteer. They immediately receive a payoff of $u_1 = -c_i$ if they volunteer and $u_1 = 0$ otherwise.
3. $t = 2$: The individual receives a payoff of $u_2 = b_i$ if they volunteered in period 1 and $u_2 = 0$ otherwise. They also receive the payoff from the commitment device or unconditional payment.

The individual has quasi-hyperbolic preferences over intertemporal payoffs and makes decisions at time $t$ that maximize $U_t = u_t + E_t[\sum_{\tau=t+1}^{T} \beta_i \delta_i^{\tau} u_\tau]$. The individual is potentially naive and thinks that they will make decisions at future time $t$ according to

$\hat{U}_t = u_t + E_t[\sum_{\tau=t+1}^{T} \hat{\beta}_i \delta_i^{\tau} u_\tau]$. Here, I borrow the notation from O'Donoghue and Rabin (2001) in which $\beta_i$ refers to the actual degree to which individual $i$'s preferences are time consistent while $\hat{\beta}_i$ refers to individual $i$'s *perceived* degree of intertemporal consistency. Assume that the cost distribution $F_i$ is uniform over $[0, \bar{c}_i]$.[14] For simplicity, assume that $\delta_i = 1$. First, consider when $\bar{c}_i > \hat{\beta}(b_i + 25)$. In this case, we will see that the individual is not 100% certain that they will volunteer with or without the commitment device.

From the perspective of $t = 0$, the individual believes that they will volunteer in period $t = 1$ if $c_i \leq \hat{\beta}_i b_i$. This implies that they believe their probability of volunteering in the absence of a commitment device is

$$\pi_{i,no} = F_i(\hat{\beta}_i b_i) = \frac{\hat{\beta}_i b_i}{\bar{c}_i}. \tag{3}$$

A commitment device in this model is a contract that pays \$25 if the individual volunteers and 0 otherwise. If they choose the commitment device, they anticipate that they will volunteer if $c_i \leq \hat{\beta}_i * (b_i + 25)$, increasing their perceived probability of volunteering to $\pi_{i,yes} = F(\hat{\beta}_i * (b_i + 25)) = \pi_{i,no} + \frac{\hat{\beta}_i * 25}{\bar{c}_i}$. If they choose the unconditional payment, their perceived probability of volunteering is simply $\pi_{i,no}$. From the perspective of $t = 0$, the individual's anticipated expected utility if they choose the commitment device is:

$$\begin{aligned} U_{0,yes} &= \beta_i * [\pi_{i,yes} * (b_i - E[c|\text{commit, vol}]) + \pi_{i,yes} * 25] \\ &= \beta_i * [(\pi_{i,no} + \frac{\hat{\beta}_i * 25}{\bar{c}_i}) * (b_i - \frac{\hat{\beta}_i * (b_i + 25)}{2}) + (\pi_{i,no} + \frac{\hat{\beta}_i * 25}{\bar{c}_i}) * 25] \end{aligned}$$

Expected utility from choosing an unconditional payment of $p$ is:

$$U_{0,no}(p) = \beta_i * [\pi_{i,no} * (b_i - E[c|\text{no commit, vol}]) + p] = \hat{\beta} * [\pi_{i,no} * (b_i - \frac{\hat{\beta}_i b_i}{2}) + p]$$

Define an individual's WTP for commitment as the amount $p$ that makes the individual indifferent between a \$25 commitment device and a \$$p$ unconditional payment.

$$wtp_i = \underbrace{(\pi_{i,yes} - \pi_{i,no}) * (b_i - \frac{\hat{\beta}_i * b_i}{2})}_{\text{Additional utility from increased probability of volunteering}} + \underbrace{(\pi_{i,yes}) * (25 - \frac{\hat{\beta}_i * 25}{2})}_{\text{Expected payout less additional expected cost}} \tag{4}$$

**Proposition 1.** *$wtp_i$ is strictly increasing in $\hat{\beta}_i$ if and only if the individual is not 100% certain that they will volunteer, with or without commitment ($\bar{c}_i > \hat{\beta}_i(b_i + 25)$).*[15]

If $\hat{\beta}_i b_i < \bar{c}_i \leq \hat{\beta}_i(b_i + 25)$, then individuals believe that they are not certain to volunteer, unless they have commitment. It is still the case that $\pi_{i,no} = \frac{\hat{\beta}_i b_i}{\bar{c}_i}$. However, in this case, individuals believe they will volunteer with certainty with commitment, so $\pi_{i,yes} = 1$. The individual's WTP for the \$25 commitment device is then:

$$wtp_i = \underbrace{(1 - \pi_{i,no})(b_i - \frac{\hat{\beta} b_i}{2})}_{\text{Additional utility from increased probability of volunteering}} + \underbrace{25}_{\text{Expected payout}} - \underbrace{\frac{\bar{c}_i - \hat{\beta}_i b_i}{2}}_{\text{Additional expected cost}} \tag{5}$$

**Proposition 2.** *$wtp_i$ is strictly decreasing in $\hat{\beta}_i$ if and only if the individual believes that they are certain to volunteer with commitment but not without ($\hat{\beta}_i b_i < \bar{c}_i \leq \hat{\beta}(b_i + 25)$).*

In the trivial case in which $\bar{c}_i \leq \hat{\beta} b_i$, the individual believes that they will volunteer with certainty, whether they have the commitment device or not. Here, WTP for commitment is simply \$25.

**Proposition 3.** *If the individual believes that they are certain to volunteer with or without commitment ($\bar{c}_i \leq \hat{\beta}_i b_i$), $wtp_i = 25$; individuals are indifferent towards pure commitment at a price of \$25.*

Propositions 1–3 characterize the WTP function for all values of $\hat{\beta}_i$ and a fixed valuation of volunteering and cost distribution. Given this characterization, we can use the sign of the derivative of WTP with respect to beliefs about probability of volunteering without a commitment device to learn about individuals' beliefs about their probability of volunteering with a commitment contract. Fig. 4 below plots WTP as a function of $\hat{\beta}_i$ for different values of $\bar{c}_i$. Notably, volunteering has a positive expected value for all of the parameterizations plotted. Pure commitment demand is represented by regions in which WTP $\geq 25$.

Lastly, it is interesting to note that there exist some individuals for whom volunteering holds a positive expected value, yet they would never demand pure commitment.

**Proposition 4.** *If $\frac{2v(b_i+25)}{2b_i+25} \leq \bar{c}_i \leq 2b_i$, the individual has a positive expected value of volunteering, yet they would not demand pure commitment for all $\hat{\beta}_i \in [0, 1]$. Furthermore, the range of maximum costs is increasing in $b_i$.*

For these individuals, no information intervention could bring them to demand commitment.

[14] This is a strong assumption about the shape of the cost distribution, but given the flexibility provided by the freedom of $\bar{c}_i$ and $b_i$, this allows for any combination of unincentivized and incentivized probabilities of volunteering.

[15] Proofs of all propositions can be found in the appendix.

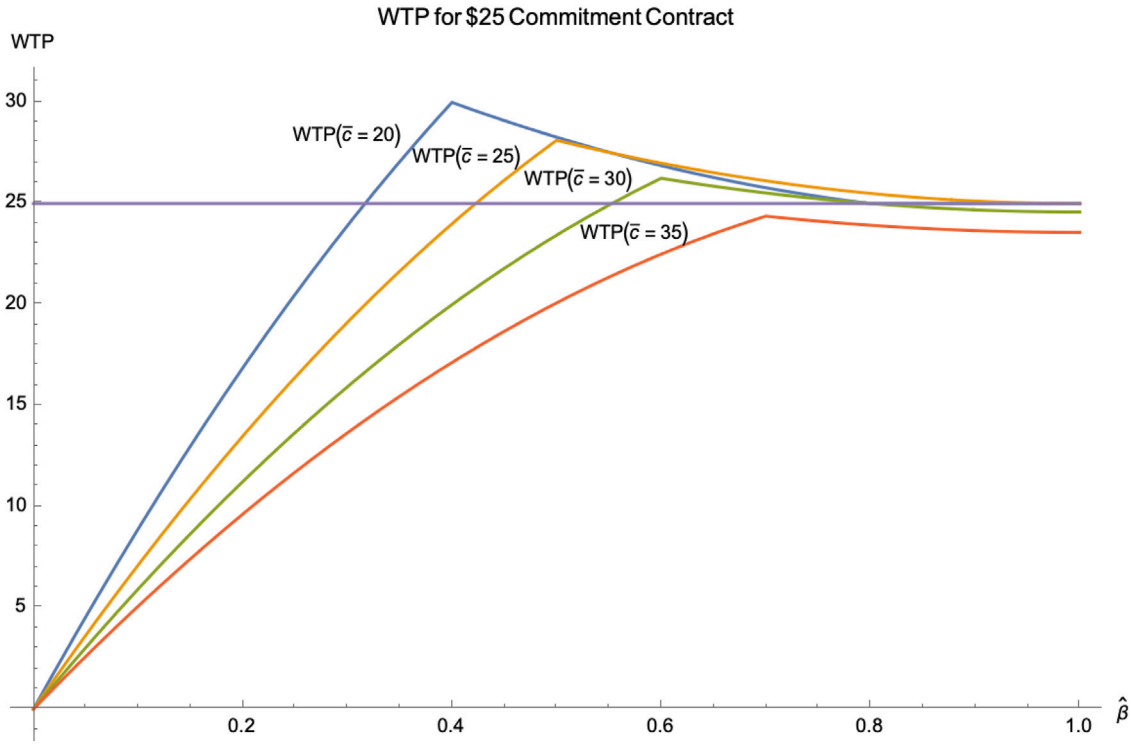



**Fig. 4.** WTP for a $25 commitment device as a function of $\hat{\beta}_i$ when $b_i = 25$ and $c_i \in \{20, 25, 30, 35\}$.

### 4.2. Discussion

Using this model as a lens to view the experimental results, Proposition 1 is useful for interpreting subjects' low commitment demand. Willingness to pay for commitment is positively correlated with beliefs about probability of follow-through. Proposition 1 tells us that this correlation will hold if and only if subjects were aware that they might fail to volunteer, even with a commitment device.

For most subjects, we can rule out the low cost/high value assumptions of Proposition 3, as we measure subjects' beliefs about their probability of volunteering in the absence of commitment. Only 21 (8%) indicate that they are certain they will volunteer, and of these, the majority have a WTP between $24 and $25 for a $25 commitment device, as Proposition 3 predicts.

The low take-up of pure commitment, the positive correlation between beliefs and WTP, and the estimated effects of information on subsidized and pure commitment (see Table 1) all suggest that most subjects are aware they might fail. While the left-censoring of the WTP measurements makes the effect of $\hat{\beta}$ on WTP difficult to estimate, we can see that the signal had a significantly positive effect on demand for commitment at a price of $24. This indicates that, on the margin, lowering subjects beliefs about their probability of volunteering reduced their WTP. This evidence is consistent with Proposition 1 in that WTP is increasing in $\hat{\beta}$. The proposition predicts that these subjects believed that they were uncertain to volunteer, even with the commitment contract. Results from the follow-up survey support this prediction. All subjects who did not have a commitment device and did not volunteer were asked, "Do you think you would have volunteered if you would have received a $25 payment conditional on you volunteering?" Only 32 of the 171 (18.7%) subjects responded "Yes" or "Definitely Yes", indicating that most of them thought it was possible that they would fail to follow-through even with a commitment contract.

Given this environment, pure commitment devices appear to be a poorly targeted intervention to improve volunteerism. The lack of commitment demand in this case is not likely naivete, but rather individuals correctly realizing that they may still fail to follow through, even in the presence of commitment. This possibility of failing with a commitment device echos the findings of John (2020) and Bai et al. (2021). However, given the observed effect of beliefs on demand, the predictions of Proposition 1, and the results of the follow-up survey, it appears that subjects were aware of this possibility of failure with commitment, driving the low demand.

Studying commitment demand in the novel setting of volunteerism presents trade-offs in interpreting the data on take-up. On one hand, it expands the base of evidence on commitment to a new setting in which intertemporal inconsistency could plausibly play a prominent role. On the other hand, the prosocial nature of the setting introduces possible confounding factors in subjects' low demand for commitment. It is possible, for example, that subjects were concerned about monetary payments "crowding out" their intrinsic motivation to volunteer (Frey and Oberholzer-Gee, 1997; Bénabou and Tirole, 2003). More simply, it is possible that

individuals' stated desire to volunteer might be exaggerated relative to their true desire to volunteer. While such explanations could explain the low level of overall commitment take-up, they are unlikely to influence the key result of this study: that commitment demand is increasing in subjects perceived probability of following through. Anticipated crowding out could lower an individual's willingness to pay for a commitment device, but this (possible) decline should be orthogonal to (and should not interact with) the treatment effect induced by learning that they are more or less likely to volunteer.

The prosociality of the setting also raises the possibility that the information treatment gave subjects "moral wiggle room" to not volunteer or commit to volunteer (Dana et al., 2007).[16] If this were the case, one should expect the information treatment to have an effect on volunteering behavior. To address this concern, I estimate Eq. (1) but replace the dependent variable with a binary variable indicating whether or not the individual volunteered (Appendix Table 5 reports the results of this regression and those of a similar probit and logit regression). The estimated coefficients on the information treatment and the signal shown to those in the information treatment (the interaction of the information treatment and the model predicted likelihood of volunteering) are insignificant. A simple comparison of the information treatment group's and the control group's volunteering rates also shows no significant difference (see Appendix Table 4). Therefore, it appears that the information treatment affected subjects' demand for commitment but not their actual volunteering behavior, a pattern that seems inconsistent with subjects using the information treatment as an excuse to not volunteer.

The overall level of commitment demand (8% for pure commitment, 26% for subsidized), is on the low end of estimates found in other settings. Among the 33 studies surveyed in Carrera et al. (2022), demand in this setting would be the 3rd lowest, and would be the lowest among similar "penalty based" contracts. Among these studies, the average take up for penalty based contracts similar to the one studied here is 37%. It is possible that some of the above hypotheses on the effects of prosociality could be lowering overall demand. However, the relationship between beliefs about following through and demand indicate that the fear of failure is also playing a significant role. A recent study by Bao and Leibbrandt (2024) finds that a similarly low 13% of subjects take up a commitment device in a puzzle solving tournament. They find that more risk averse subjects were less likely to take up the commitment device, indicating that a similar fear of uncertainty and failure with the commitment device, may drive low commitment take-up in other settings. While the agents in my model are risk neutral, risk aversion would strengthen the channel studied and would predict that finding of Bao and Leibbrandt (2024).

Given the negligible effects of offering commitment on volunteering follow-through, it is worth exploring other approaches to increasing volunteerism. The results of Section 3.3 indicate that monetary incentives increase volunteering, echoing the findings of Carpenter and Myers (2010) and Lacetera et al. (2014) (although the former finds that this effect is moderated by image concerns). Given the very definition of volunteering, monetary incentives may not be a desirable method of increasing participation. Moseley et al. (2018) find that social information on volunteering time does not impact volunteerism. Yang (2017) and McCarty et al. (2018) both find that mandated volunteering is ineffective at increasing future volunteerism. Brown et al. (2019) and Lilley and Slonim (2014) show that the warm glow from volunteering is greater than that of donating money. Carpenter and Myers (2010) find that concern for image and reputation are positively correlated with volunteering. Highlighting the potential warm glow and reputational effects of volunteering may therefore be fruitful avenues for increasing volunteering.

While the above results on the connection between beliefs about the probability of failure and commitment demand are specific to the setting studied in this paper, they provide additional evidence that the settings under which commitment demand arises are rare. It is possible that a similar fear of failure is suppressing demand for commitment in other settings as well. This possibility of failure could stem from two sources: the ineffectiveness of the commitment device or the inherent uncertainty of the setting. John (2020) finds that many people choose commitment levels that are too weak and thus do not provides sufficient incentive to change behavior. It is possible that subjects perceived the $25 incentive in this study as too small, leading to higher perceived probability of failure, and low overall levels of commitment demand. Laibson (2015) shows that sufficient uncertainty can reduce demand for commitment. Lack of overall demand may therefore stem from considerable volatility in the perceived value of students' time. Practitioners considering commitment as an intervention should therefore focus on settings in which their commitment devices will ensure follow-through.

## 5. Conclusion

While laboratory studies have shown modest demand for commitment, there appear to be few natural settings in which they arise. This study suggests that this lack of demand is not a function of naivete on the part of decision makers, but rather the correct realization that failure to follow-through is possible, even with the offered commitment contract. For students who have indicated an interest in volunteering with non-profit organizations, the possibility of failure to follow through, even with a commitment device, outweighs the increased probability of volunteering that a commitment device provides. For this reason, voluntary commitment devices are unlikely to improve volunteerism or student welfare.

## Declaration of competing interest

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This study was approved by the IRB at Boston College (Protocol Number: 21.147.01).

There are no potential conflicts of interest regarding this paper.

[16] Thank you to an anonymous referee for this suggestion.

## Data availability

Data will be made available on request.

## Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jebo.2024.106756.

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